TÉCNICO LISBOA

Data Analysis and Integration

Portugal's Economic and Human Development - A Century of Transformation

Project 26/27 – Group 20
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Task 1 – Staging and Data Profiling

Staging database, load pipeline, exploratory queries and data dictionaries

Staging schema with precise data types

```sql
CREATE TABLE stg_maddison (

  country_iso3    CHAR(3)        NOT NULL,
  country_name    VARCHAR(100)   NOT NULL,
  indicator_code  VARCHAR(10)    NOT NULL,
  indicator_name  VARCHAR(100)   NOT NULL,
  year            SMALLINT       NOT NULL,
  value           DECIMAL(22,6)  NULL,

  PRIMARY KEY (country_iso3, indicator_code, year)
) ENGINE = InnoDB;

CREATE TABLE stg_wdi (

  country_iso3    CHAR(3)        NOT NULL,
  country_name    VARCHAR(100)   NOT NULL,
  indicator_code  VARCHAR(30)    NOT NULL,
  indicator_name  VARCHAR(200)   NOT NULL,
  year            SMALLINT       NOT NULL,
  value           DECIMAL(22,6)  NULL,

  PRIMARY KEY (country_iso3, indicator_code, year)
) ENGINE = InnoDB;

CREATE TABLE stg_events (

  event_id        INT            NOT NULL AUTO_INCREMENT,
  year            SMALLINT       NOT NULL,
  country_iso3    CHAR(3)        NOT NULL,
  country_name    VARCHAR(100)   NOT NULL,
  event           VARCHAR(500)   NOT NULL,
  category        VARCHAR(50)    NOT NULL,
  economic_impact VARCHAR(100)   NULL,

  PRIMARY KEY (event_id)
) ENGINE = InnoDB;
```

Maddison and Wdi use the same composite primary key, each row is one value per country, indicator and year, so country and the indicator alone repeat every year

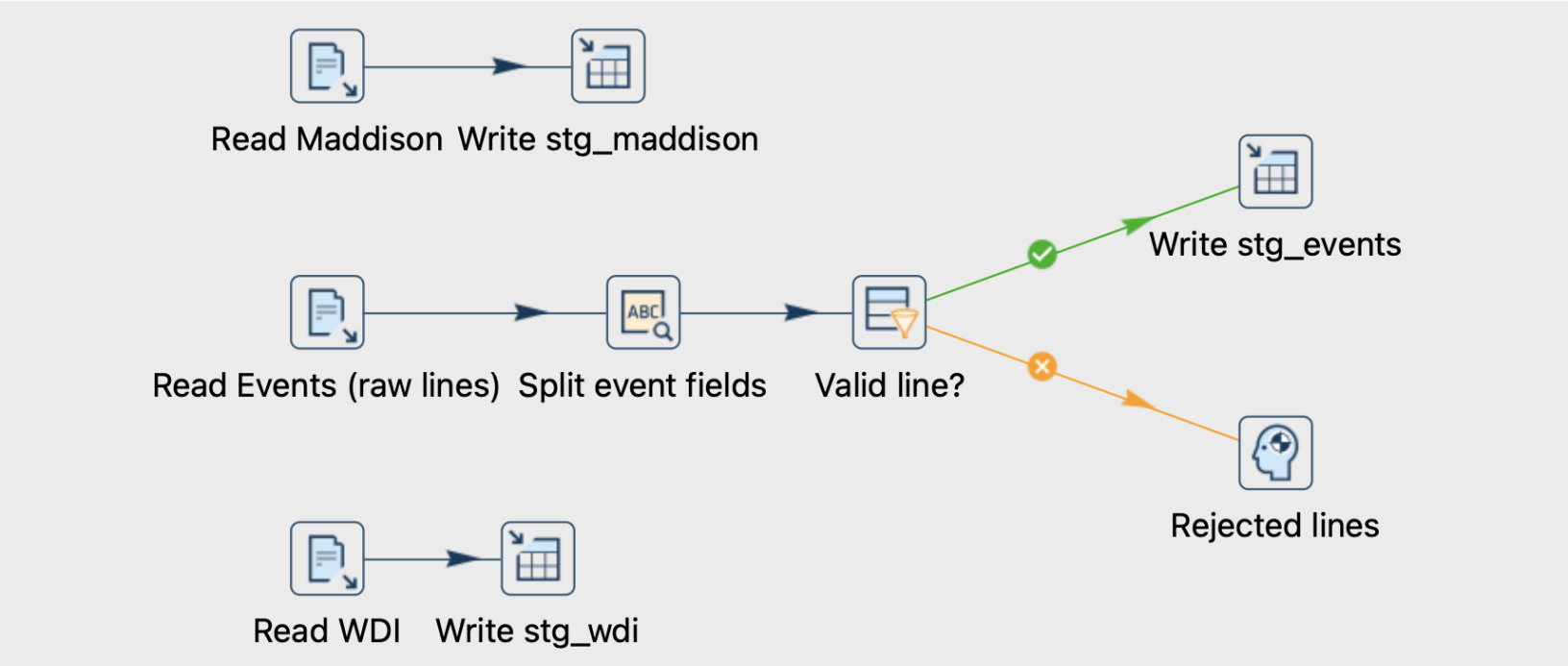
Events has no natural key so it was necessary to create a surrogate key

country_iso3	country_name	indicator_co...	indicator_name	year	value
BEL	Belgium	gdp	GDP (2011 international $)	1900	39957893000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1901	40316328000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1902	41141880000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1903	42072961000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1904	43160826000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1905	44398900000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1906	45318952000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1907	45994584000.000000
BEL	Belgium	gdp	GDP (2011 international $)	1908	46459559000.000000
BEL	Belgium	g			
BEL	Belgium	g			
BEL	Belgium	g			
BEL	Belgium	g			
BEL	Belgium	g			
BEL	Belgium	g			
BEL	Belgium	g			

country_iso3	country_name	indicator_code	indicator_name	year	value
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1970	44.956276
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1971	43.767463
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1972	44.137278
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1973	48.031944
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1974	52.893358
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1975	45.916037
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1976	48.554383
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1977	47.507250
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1978	45.761644
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1979	49.922438
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1980	50.117280
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1981	54.063522
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1982	58.103312
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1983	60.395627
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1984	64.412352
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1985	62.454327
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1986	57.283050
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1987	55.867763
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1988	59.408083
BEL	Belgium	NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	1989	63.566749

event_id	year	country_iso3	country_name	event	category	economic_impact
1	1900	PRT	Portugal	Economy overwhelmingly agrarian; illiteracy above 70%; one of poorest Western European countries	Economy	Stagnation
2	1902	PRT	Portugal	Second state bankruptcy (first was 1892); public finances collapse	Financial	Debt crisis
3	1908	PRT	Portugal	Assassination of King Carlos I and Crown Prince Luís Filipe (1 February)	Political	Regime crisis
4	1910	PRT	Portugal	Proclamation of the Republic (5 October); end of the monarchy; separation of Church and State	Political	Regime change
5	1911	PRT	Portugal	Creation of University of Lisbon and University of Porto; education expansion	Social	Education reform
6	1916	PRT	Portugal	Portugal enters World War I on the Allied side (March)	War	Military expenditure; inflation
7	1917	PRT	Portugal	Battle of La Lys; devastating losses for Portuguese Expeditionary Corps (April)	War	Human cost
8	1918	PRT	Portugal	End of WWI; Spanish flu pandemic devastates Portugal	War/Health	Population decline
9	1920	PRT	Portugal	Extreme political instability; rotating governments; chronic inflation and fiscal deficits	Political	Economic chaos
10	1924	PRT	Portugal	Banco de Portugal scandal; fraudulent banknote printing by Alves dos Reis	Financial	Banking crisis
11	1926	PRT	Portugal	Military coup (28 May); end of the First Republic; beginning of military dictatorship	Political	Regime change
12	1928	PRT	Portugal	Salazar appointed Minister of Finance; imposes fiscal austerity and balanced budgets	Economy	Fiscal stabilization
13	1931	PRT	Portugal	Condicionamento Industrial introduced; state licensing controls all industrial investment	Economy	State-controlled industry
14	1933	PRT	Portugal	Constitution of the Estado Novo; corporatist authoritarian regime formally established	Political	Regime formalized
15	1935	PRT	Portugal	Lei de Reconstituição Económica; 15-year public investment plan; quadruples annual public investment	Economy	Infrastructure investment plan

Load pipeline (one branch per source)



Maddison

CSV file input → Table output (stg_maddison)

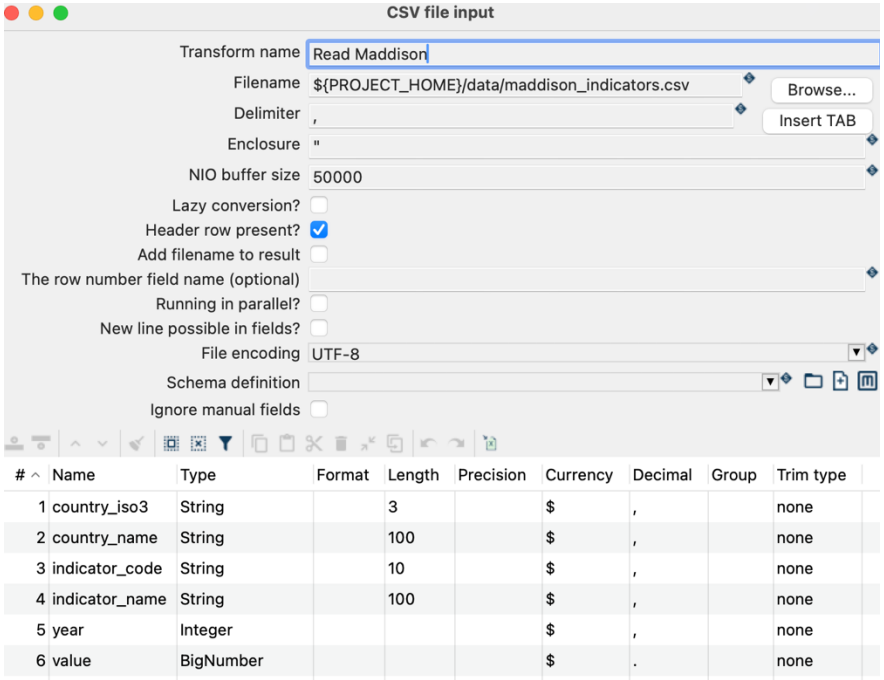
Events

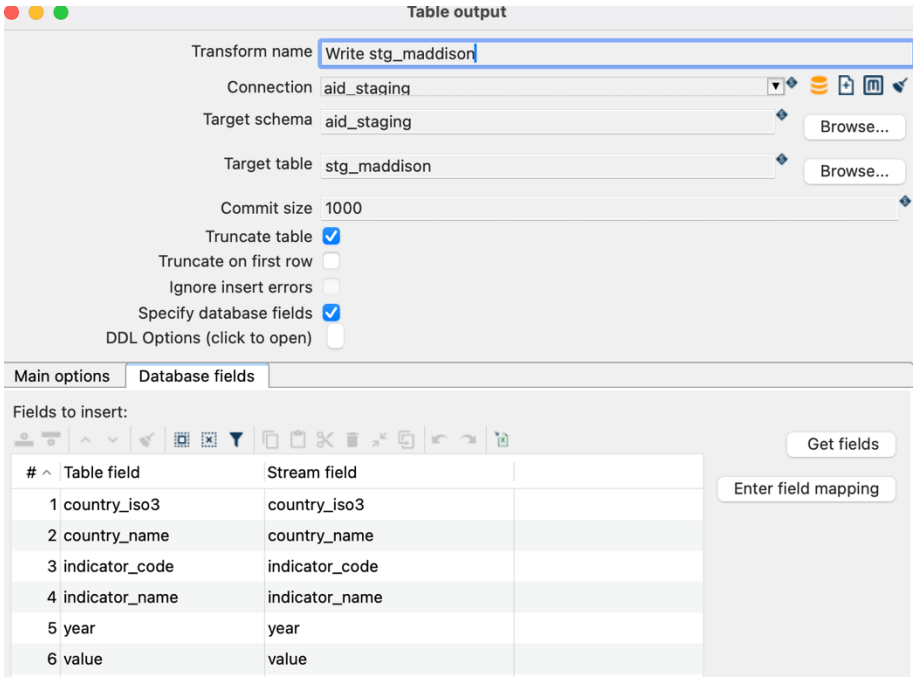
CSV file input → Regex evaluation → Filter rows → Table output

WDI

CSV file input → Table output (stg_wdi)

Key transform configurations





Path with ${PROJECT_HOME}, not an absolute path

year is Integer, value is BigNumber

Truncate table, the load can be re-run

Explicit mapping; event_id left to MySQL

An unprotected comma

1990,IRL,Ireland,Celtic Tiger boom begins; strong FDI, IT and pharmaceuticals growth through the 1990s,Economy,High growth

We noted the presence of an unprotected comma in line 362 of the events csv file, which split that row into 7 fields instead of 6. Instead of editing the source file, the load pipeline reads each line and splits it with a regular expression that allows commas, only in the event field

Read the whole line (delimiter |) → **Regex split into 6 fields** → **Filter is_valid = Y** → **Table output stg_events**

```
^(\d{4}),([A-Z]{3}),([^,]*),(.*),([^,]*),([^,]*)$
```

350	1900	IRL	Ireland	Ireland is part of the United Kingdom; largely agrarian with emigration after the Famine	Political	Colonial status
351	1916	IRL	Ireland	Easter Rising in Dublin; republican rebellion against British rule	Political	Independence path
352	1919	IRL	Ireland	War of Independence begins; guerrilla conflict with British forces	War	Political violence
353	1921	IRL	Ireland	Anglo-Irish Treaty; Irish Free State created (1922); partition of Northern Ireland	Political	State creation
354	1922	IRL	Ireland	Irish Civil War over the Treaty (1922-1923)	War	Civil war
355	1932	IRL	Ireland	De Valera government; economic war with Britain over land annuities	Economy	Trade conflict
356	1937	IRL	Ireland	New constitution; the state is named Ireland (Éire)	Political	Constitutional reform
357	1949	IRL	Ireland	Republic of Ireland Act; leaves the British Commonwealth	Political	Republic declared
358	1958	IRL	Ireland	First Programme for Economic Expansion; shift from protectionism toward FDI	Economy	Policy turning point
359	1973	IRL	Ireland	Ireland joins the EEC (1 January) with the UK and Denmark	Integration	European integration
360	1979	IRL	Ireland	Ireland joins the European Monetary System; breaks sterling parity	Integration	Monetary autonomy
361	1990	IRL	Ireland	Celtic Tiger boom begins; strong FDI, IT and pharmaceuticals growth through the 1990s	Economy	High growth
362	1999	IRL	Ireland	Ireland adopts the Euro; founding Eurozone member	Integration	Monetary union
363	2008	IRL	Ireland	Banking collapse and property crash; EU-IMF bailout in 2010	Financial	Bailout
364	2013	IRL	Ireland	Ireland exits the EU-IMF programme	Financial	Recovery begins
365	2015	IRL	Ireland	GDP jumps after large multinational inversions; GDP becomes a poor welfare measure versus GNI	Economy	Statistical distortion
366	2020	IRL	Ireland	COVID-19; GDP holds up better than GNI because of multinational output	Economy	Pandemic shock
367	1900	HUN	Hungary	Kingdom of Hungary inside Austria-Hungary; mixed agrarian and industrial economy	Political	Imperial status
368	1918	HUN	Hungary	Aster Revolution; end of the dual monarchy after World War I	Political	Regime change
369	1920	HUN	Hungary	Treaty of Trianon; Hungary loses about two thirds of its territory and population	Political	Territorial loss
370	1929	HUN	Hungary	Great Depression; agrarian collapse and financial crisis	Economy	GDP decline
371	1941	HUN	Hungary	Hungary enters World War II as an Axis ally	War	War economy

All rows loaded into staging

4,349
stg_maddison
4,350 lines incl. header

387
stg_events
388 lines incl. header

11,371
stg_wdi
11,372 lines incl. header

#	Transform Name	Copy	Input	Read	Written	Output	Updated	Rejected	Errors	Buffers Input	Buffers Output	Duration	Speed	Status
1	Read Maddison	0	4,350	0	4,349	0	0	0	0	0	0	0.077"	45,312	Finished
2	Write stg_maddison	0	0	4,349	4,349	4,349	0	0	0	0	0	0.357"	11,566	Finished
3	Read Events (raw lines)	0	388	0	387	0	0	0	0	0	0	0.009"	13,379	Finished
4	Split event fields	0	0	387	387	0	0	0	0					
5	Valid line?	0	0	387	387	0	0	0	0					
6	Write stg_events	0	0	387	387	387	0	0	0					
7	Rejected lines	0	0	0	0	0	0	0	0					
8	Read WDI	0	11,372	0	11,371	0	0	0	0					
9	Write stg_wdi	0	0	11,371	11,371	11,371	0	0	0					

source_table	n_rows
stg_maddison	4349
stg_wdi	11371
stg_events	387

Profiling - six sections of queries (overview)

Section	Question	Key fact found
1. Row count per table	Which countries are in each source?	EU missing only in Maddison
2. Year ranges	Which years per source and country?	1900–2025 overall but some gaps in the East
3. WDI coverage	Since when is each indicator available?	Sectors from 1995 and employment from 1991
4. Missing values	Where is data missing?	0 NULLs missing data = absent rows
5. Data integrity	Are there duplicates? Consistent values?	0 duplicates gdp = gdppc × pop × 1000
6. Historical events	How are events distributed?	Portugal 125 of 387

Profiling - six sections of queries (query screenshots)

country_iso3	indicator_co...	first_year	last_year	n_years	missing_inside
BEL	gdp	1900	2022	123	0
BEL	gdppc	1900	2022	123	0
BEL	pop	1900	2022	123	0
CZE	gdp	1970	2022	53	0
CZE	gdppc	1970	2022	53	0
CZE	pop	1970	2022	53	0
DEU	gdp	1900	2022	123	0
DEU	gdppc	1900	2022	123	0
DEU	pop	1900	2022	123	0
ESP	gdp	1900	2022	123	0
ESP	gdppc	1900	2022	123	0
ESP	pop	1900	2022	123	0
EST	gdp	1973	2022	44	6
EST	gdppc	1973	2022	44	6
EST	pop	1973	2022	44	6
FRA	gdp	1900	2022	123	0
FRA	gdppc	1900	2022	123	0
FRA	pop	1900	2022	123	0
GRC	gdp	1900	2022	123	0
GRC	gdppc	1900	2022	123	0
GRC	pop	1900	2022	123	0
HUN	gdp	1900	2022	100	23
HUN	gdppc	1900	2022	100	23
HUN	pop	1900	2022	100	23
IRL	gdp	1913	2022	103	7
IRL	gdppc	1913	2022	103	7
IRL	pop	1913	2022	103	7
ITA	gdp	1900	2022	123	0
ITA	gdppc	1900	2022	123	0
ITA	pop	1900	2022	123	0
LTU	gdp	1973	2022	44	6
LTU	gdppc	1973	2022	44	6
LTU	pop	1973	2022	44	6
LVA	gdp	1973	2022	44	6
LVA	gdppc	1973	2022	44	6
LVA	pop	1973	2022	44	6
POL	gdp	1900	2022	89	34
POL	gdppc	1900	2022	89	34
POL	pop	1900	2022	89	34
PRT	gdp	1900	2022	123	0
PRT	gdppc	1900	2022	123	0
PRT	pop	1900	2022	123	0
ROU	gdp	1900	2022	106	17

indicator_code	PRT	ESP	FRA	DEU	BEL	ITA	IRL	GRC	POL	CZE	HUN	ROU	EST	LVA
NE.EXP.GNFS.ZS	56	56	66	56	56	56	56	66	31	36	35	36	33	31
NE.IMP.GNFS.ZS	56	56	66	56	56	56	56	66	31	36	35	36	33	31
NV.AGR.TOTL.CD	31	31	66	35	31	36	31	31	31	33	31	36	31	31
NV.AGR.TOTL.ZS	31	31	66	35	31	36	31	31	31	33	31	36	31	31
NV.IND.MANF.CD	31	31	66	35	31	36	31	31	31	33	31	35	31	31
NV.IND.MANF.ZS	31	31	66	35	31	36	31	31	31	33	31	35	31	31
NV.IND.TOTL.CD	31	31	66	35	31	36	31	31	31	33	31	36	31	31
NV.IND.TOTL.ZS	31	31	66	35	31	36	31	31	31	33	31	36	31	31
NV.SRV.TOTL.CD	31	31	66	35	31	36	31	31	31	33	31	36	31	31
NV.SRV.TOTL.ZS	31	31	66	35	31	36	31	31	31	33	31	36	31	31
NY.GDP.MKTP.K...	65	65	65	65	65	65	65	65	35	35	65	35	35	35
NY.GNP.PCAP.CD	64	64	64	64	64	64	64	64	34	34	56	34	31	29

country_iso3	n_partial_years	first_partial_y...	last_partial_y...
ROU	17	1901	1919

country_iso3	avg_value_added_s...	min_value_added_s...	max_value_added_s...	avg_employment_s...
BEL	89.6	89.2	90.6	100.0
CZE	90.7	89.7	91.9	100.0
DEU	89.9	89.2	91.3	
ESP	91.2	89.4	93.7	
EST	87.9	86.1	89.9	
EUU	89.5	89.0	90.4	
FRA	88.9	87.2	90.1	
GRC	88.4	86.2	90.9	

decade	n_events_all	n_events_prt
1900	16	3
1910	26	5
1920	28	4
1930	20	4
1940	47	7
1950	2	
1960	2	
1970	3	
1980	2	
1990	5	

country_iso3	year	n_events
PRT	1961	5
PRT	1973	4
PRT	1975	4
PRT	1991	4
PRT	1954	3
PRT	1964	3
PRT	1998	3
PRT	2004	3
PRT	1958	2
PRT	1959	2
PRT	1960	2
T	1967	2
T	1972	2
T	1974	2
T	1976	2
T	1977	2
T	1978	2
T	1986	2
T	1993	2
T	1995	2
T	2000	2
T	2001	2
T	2002	2
T	2014	2

source_table	n_duplicate_ke...
stg_maddison	0
stg_wdi	0
stg_event	

category	n_
War/Health	1
War/Industry	1
Integration/Industry	1
Infrastructure/Economy	1
Integration/Security	1

n_distinct_impacts
212

source	indicator_c			outside_0_1...	
maddison	gdp				
maddison	gdppc	853.242430	60257.082000	0	0
maddison	pop	1311.859994	83811.312444	0	0
wdi	NE.EXP.GNFS.ZS	7.617289	143.984602	0	16
wdi	NE.IMP.GNFS.ZS	12.401541	126.428264	0	6
wdi	NV.AGR.TOTL.CD	186001573.564123	346416185370.951000	0	0
wdi	NV.AGR.TOTL.ZS	0.557956	21.809069	0	0
wdi	NV.IND.MANF.CD	764623131.392604	3031607091312.370000	0	0
wdi	NV.IND.MANF.ZS	7.780487	37.098038	0	0
wdi	NV.IND.TOTL.CD	1232978756.884340	4715981808430.980000	0	0
wdi	NV.IND.TOTL.ZS	13.413957	45.879473	0	0
wdi	NV.SRV.TOTL.CD	2568733280.881200	13996637427932.4000...	0	0
wdi	NV.SRV.TOTL.ZS	24.175312	71.757544	0	0
wdi	NY.GDP.MKTP.K...	-32.118570	42.410414	125	0
wdi	NY.GNP.PCAP.CD	430.000000	87360.000000	0	0
wdi	SL.AGR.EMPL.ZS	0.862861	33.605436	0	0
wdi	SL.IND.EMPL.ZS	15.018662	47.845953	0	0
wdi	SL.SRV.EMPL.ZS	30.339969	79.768583	0	0
wdi	SP.DYN.LE00.IN	62.666512	83.951220	0	0
wdi	SP.URB.TOTL.IN...	30.841300	87.672608	0	0

economic_impact	n_events
Pandemic shock	15
Monetary union	12
Regime change	11
European integration	8
State creation	7
Recession	6
GDP decline	6
Authoritarian rule	6
Repression	6
Banking crisis	5
Growth shock	5
War economy	5
GDP collapse	5
Democratic transition	5
Western integration	5
Occupation	5
Territorial loss	4
Energy expansion	4
Recovery begins	4
GDP collapse; dev...	4
EU membership	4
Loss of sovereignty	4
Colonial status	4
Debt crisis	3
Austerity	3
Event investment	3
Bailout	3
Policy shift	3
GDP collapse; occu...	3
Recovery starts	3
Social unrest	3
Communist regime...	3
Transition shock	3
Sovereignty confir...	3
Fate sealed	3
Soviet rule restored	3
National movement	3
State restored	3
Currency reform	3
EU and NATO mem...	3
Severe recession	3
Road modernization	2
Western alignment	2
Planned industrializ...	2
Demographic shock	2

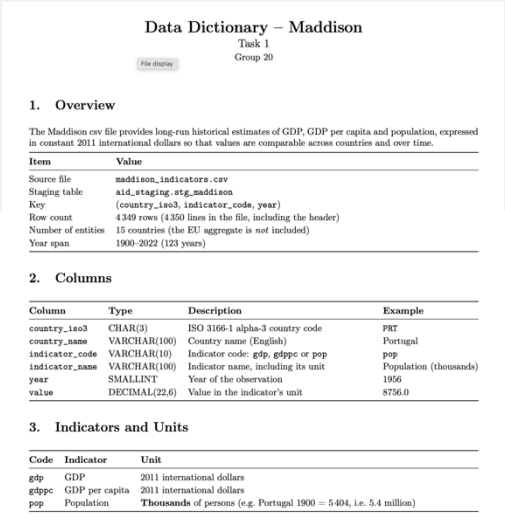
Data dictionaries

Maddison

4,349 rows · 15 countries · 1900–2022

Particularities and data quality:

- EU missing
- Missing data as absent rows
- Romania – misalign indicators



Data Dictionary – Maddison
Task 1
Group 20

1. Overview

The Maddison csv file provides long-run historical estimates of GDP, GDP per capita and population, expressed in constant 2011 international dollars so that values are comparable across countries and over time.

Item	Value
Source file	maddison_indicators.csv
Staging table	aid_staging.stg_maddison
Key	(country_iso3, indicator_code, year)
Row count	4 349 rows (4 350 lines in the file, including the header)
Number of entities	15 countries (the EU aggregate is *not* included)
Year span	1900–2022 (123 years)

2. Columns

Column	Type	Description	Example
country_iso3	CHAR(3)	ISO 3166-1 alpha-3 country code	PRT
country_name	VARCHAR(100)	Country name (English)	Portugal
indicator_code	VARCHAR(10)	Indicator code: gdp, gdppc or pop	pop
indicator_name	VARCHAR(100)	Indicator name, including its unit	Population (thousands)
year	SMALLINT	Year of the observation	1956
value	DECIMAL(22,6)	Value in the indicator's unit	8756.0

3. Indicators and Units

Code	Indicator	Unit
gdp	GDP	2011 international dollars
gdppc	GDP per capita	2011 international dollars
pop	Population	**Thousands** of persons (e.g. Portugal 1900 = 5 404, i.e. 5.4 million)

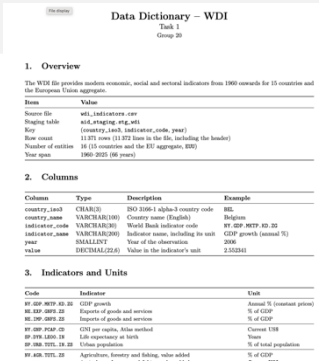
No indicator has negative values. gdp is a derived value: $gdp = gdppc \times pop \times 1000$

WDI

11,371 rows · 16 entities · 1960–2025

Particularities and data quality:

- Missing data as absent rows
- EU included
- Manufacturing is part of industry
- No employment for manufacturing
- Extreme GDP growth



Data Dictionary – WDI
Task 1
Group 20

1. Overview

The WDI file provides modern economic, social and sectoral indicators from 1960 onwards for 15 countries and the European Union aggregate.

Item	Value
Source file	wdi_indicators.csv
Staging table	aid_staging.stg_wdi
Key	(country_iso3, indicator_code, year)
Row count	11 371 rows (11 372 lines in the file, including the header)
Number of entities	16 (15 countries and the EU aggregate, EUU)
Year span	1960–2025 (66 years)

2. Columns

Column	Type	Description	Example
country_iso3	CHAR(3)	ISO 3166-1 alpha-3 country code	BEL
country_name	VARCHAR(100)	Country name (English)	Belgium
indicator_code	VARCHAR(30)	World Bank indicator code	NY.GDP.MKTP.KD.ZG
indicator_name	VARCHAR(200)	Indicator name, including its unit	GDP growth (annual %)
year	SMALLINT	Year of the observation	2006
value	DECIMAL(22,6)	Value in the indicator's unit	2.552341

3. Indicators and Units

Code	Indicator	Unit
NY.GDP.MKTP.KD.ZG	GDP growth	Annual % (constant prices)
NE.EXP.GNFS.ZS	Exports of goods and services	% of GDP
NE.IMP.GNFS.ZS	Imports of goods and services	% of GDP
NY.GDP.PCAP.CD	GNI per capita, Atlas method	Current US$
SP.DYN.LE00.IN	Life expectancy at birth	Years
SP.URB.TOTL.IN.ZS	Urban population	% of total population
NV.AGR.TOTL.ZS	Agriculture, forestry and fishing, value added	% of GDP

Events

387 events · 16 entities · 1900–2025

Particularities and data quality:

- Unprotected comma in one row
- No natural key
- Composite categories
- No missing fields
- Coverage beyond the indicators



Data Dictionary – Events
Task 1
Group 20

1. Overview

The events file provides a chronology of major economic, political, social and infrastructure events for 15 countries and the European Union aggregate, from 1900 to 2025.

Item	Value
Source file	historical_events.csv
Staging table	aid_staging.stg_events
Key	event_id (surrogate key generated on load; the file has no natural key)
Row count	387 events (388 lines in the file, including the header)
Number of entities	16 (15 countries and the EU aggregate, EUU)
Year span	1900–2025 (110 distinct years)

2. Columns

Column	Type	Description	Example
event_id	INT	Surrogate key	21
year	SMALLINT	Year of the event	1945
country_iso3	CHAR(3)	ISO 3166-1 alpha-3 country code	PRT
country_name	VARCHAR(100)	Country name (English)	Portugal
event	VARCHAR(100)	Description of the event	First motorway segment; …
category	VARCHAR(50)	Type of event	Infrastructure
economic_impact	VARCHAR(100)	Economic impact of the event	Road modernisation

3. Categories and Impacts

Category	Events
Political	102
Economy	90
Integration	40
War	39
Financial	32
Infrastructure	31